

The geometric inverse 3/2 law is proposed as a fundamental framework for the construction of virtual digital universes. The inverse 3/2 law is derived from Kepler's Third Law.

Author: Vitali Colesnicenco

Independent Researcher, London, England, United Kingdom

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Introduction:

The scope of application for a geometric model of the universe based on the inverse 3/2 law. The modeling of gravitational systems and digital universes is based on only two geometric parameters: the radius and the system time.

This approach is particularly advantageous for educational and gaming simulators, such as [Kerbal Space Program](#), where the precise configuration of orbits and planetary behavior requires simple, physically meaningful input parameters.

Construction: A vertical axis is drawn upwards from a central point. Arcs expand from this axis with radii increasing in integer steps ($r, 2r, 3r\dots$). The arc's opening angle decreases as the distance from the initial central point increases. The first arc, with a radius of r unfolds completely to form a circle

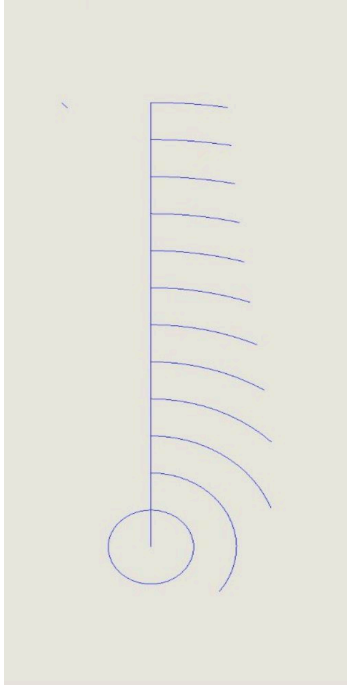
The second arc with a radius of $2r$ ($n=2$) opens to an angle of

$$\alpha = \frac{2\pi}{2^{3/2}}$$

Any other arc with a radius r_n opens to an angle of

$$\alpha_n = \frac{2\pi}{n^{3/2}}$$

The arcs can be extended to infinity, which indicates that phenomena built upon the foundation of this law are capable of interacting across infinite distances.



This is a 2D static drawing, but even at this stage, it can be observed that the square of any arc multiplied by its radius yields the same value.

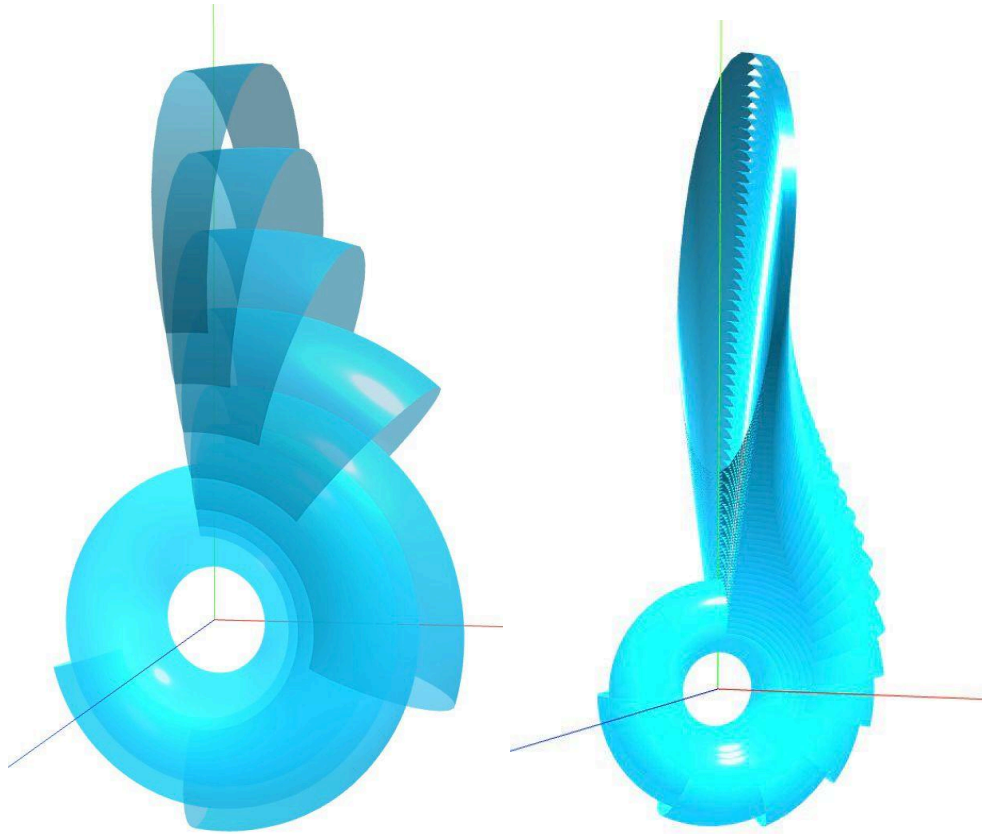
$$L_n^2 \cdot r_n = \left(\frac{2\pi \cdot nr}{n^{3/2}} \right)^2 \cdot (nr) = \left(\frac{4\pi^2 r^2}{n} \right) \cdot (nr) = 4\pi^2 r^3 = \text{const}$$

This constant can be associated with mass—it is an **inertial parameter**. The symbol of resistance to external influence is the gyroscope. A torus exhibits a gyroscopic effect. Geometrical Mass is defined as an inertial parameter where **1 m³** of torus sector volume corresponds to **585 kg** of physical mass.

To visualize this constant, I chose the volume of a torus with a tube radius equal to half of the torus radius, $r = 0.5R$

The 3D visualization consists of equal-volume torus sectors.

<https://vitalydecoder.github.io/BaseReal-OS/>

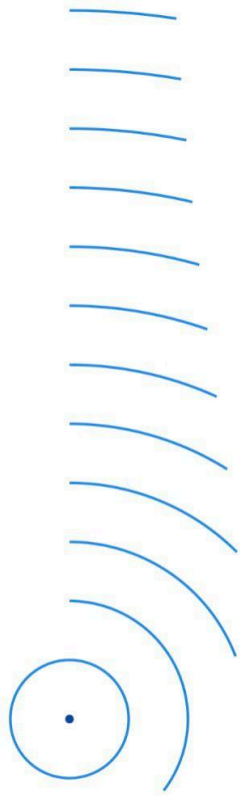


DYNAMICS. ADDING THE PERIOD

The proposed geometric model is based on the inverse 3/2 law, visually represented by a system of arcs unfolding from a central axial line. The opening angle of each arc at a height of radii is determined by the expression:

$$\alpha_n = \frac{2\pi}{n^{3/2}}$$

<https://vitalydecoder.github.io/BaseReal-OS/speed.html>



Once system time T is added, each arc shows the path a body travels at its current distance from the center during one system period. This results in a velocity distribution scheme. Angular velocity - ω . Orbital velocity - v .

$$\frac{2\pi}{T} = \omega \text{ - Angular velocity}$$

$$\frac{2\pi R}{T} = v \text{ - Orbital velocity}$$

After assigning the time period T , a second constant emerges.

The square of the angular velocity of the arc's opening multiplied by the cube of the arc's radius equals a constant. This constant can be associated with energy—it is the

μ - **gravitational parameter** of the field.

$$\mu = \omega^2 r^3 = \frac{4\pi^2}{T^2} r^3 = \frac{4\pi^2 r^3}{T^2} = \text{const}$$

Or:

$$\mu = v^2 r = \frac{4\pi^2 r^2}{T^2} r = \frac{4\pi^2 r^3}{T^2} = \text{const}$$

The intensity of this parameter manifests as centripetal acceleration, which decreases with the square of the distance from the center of the field. Thus, when the period is added, the inverse 3/2 law generates dynamics, and the inverse square law naturally follows from it. In this model, the mass of a body is determined purely geometrically:

$$M = 4\pi^2 R^3$$

It is noteworthy that the Earth's mass value calculated in this manner is very close to the result obtained by Cavendish when determining the mean density of the Earth.

*A key feature of this model is that the **gravitational parameter** and **mass** does not maintain a direct linear proportionality; rather, each object is characterized by its own fundamental period T .*

The minimum astronomical unit used is the - **kilosecond (1000sec)**.

The parameters for Earth, Mars, and the Sun are particularly well-known. These bodies have precisely measured radii, possess natural satellites (allowing for reliable determination of their radius and gravitational parameter and are sufficiently close for direct observation and research. For these three bodies, the following fundamental periods are used within the framework of the model:

Earth — $T = 5000$ seconds (5 kiloseconds), (anomaly +60 s)

Mars — $T = 6000$ seconds (6 kiloseconds)

Sun — $T = 10000$ seconds (10 kiloseconds)

The Earth anomaly of +60s is explained by a 50km radius increase, consistent with the Theia impact hypothesis, while the gravitational field remains unchanged.

In this model, mass follows the calculated field, keeping the gravitational parameter (μ) and the inertial parameter (M) entirely decoupled. This provides a natural gain of several percent for small bodies during barycenter calculations

Mutual Attraction and Newton's Third Law in Action

In the proposed geometric model, each body simultaneously serves as the source of its own gravitational field ($\mu = \frac{4\pi^2 R^3}{T^2}$) and an inertial object ($M = 4\pi^2 R^3$).

The dominant body's field indicates the direction and magnitude of acceleration to both masses (the "hammer and feather" principle is preserved). The dominant field intensity (acceleration) at the centre of the test body gives a specific acceleration. This acceleration should be divided between the bodies in inverse proportion to their masses. Thus, we automatically obtain perfect compliance with Newton's third law. The forces are equal and opposite.

Example: Mutual orbital perturbation. Earth-Venus

$\mu_{[Earth]}$ – gravitational parameter.

$$\mu_{[Earth]} = 4e14 \, m^3/s^2$$

$R = 4.2e10$ (average Earth-Venus distance at opposition / inferior conjunction)

$$a = \frac{\mu_{[Earth]}}{R^2} = \frac{4e14}{4.2e10}$$

First, we calculate the square of the distance:

$$R^2 = (4.2e10)^2 = 1.764e21 \text{ m}^2$$

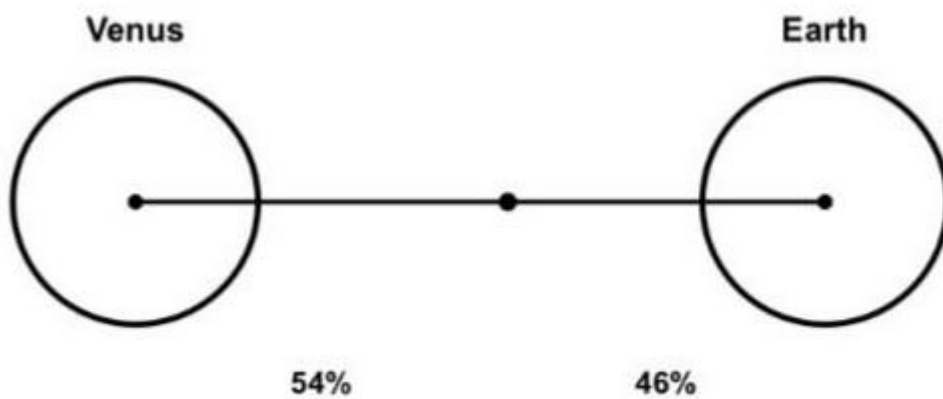
Then the acceleration:

$$a = \frac{4e14}{1.764e21} = 2.27e-7 \text{ m/s}^2$$

Total acceleration:

$$a = 2.27e-7 \text{ m/s}^2$$

(or approximately 0.000000227 m/c²)



The barycentre perfectly shows what percentage of acceleration is assigned to each.

For Venus and Earth, the calculation was performed using the specified coefficients and the mass expression in the form of: $M = 4\pi^2 r^3$

$$r(\text{Venus}) = 6.052e6 \text{ m} \text{ --- Venus radius,}$$

$$r(\text{Earth}) = 6.371e6 \text{ m} \text{ --- Earth radius,}$$

Gravitational acceleration is calculated at a distance of $4.2e10 \text{ m}$ (Earth–Venus at opposition).

$$a = 2.26757e-7 \text{ m/s}^2$$

$$F_1 = 0.54 \times a \times (4\pi^2 r^3 [\text{Venus}]) = 1.072e15 \text{ m}^4/\text{s}^2$$

$$F_2 = 0.46 \times a \times (4\pi^2 r^3 [\text{Earth}]) = 1.065e15 \text{ m}^4/\text{s}^2$$

where:

With a more accurate calculation of the Earth–Venus system barycentre, the values of F_1 and F_2 become equal.

In this geometric model, gravity works as a simple algorithm for distributing a single acceleration. Instead of calculating forces for each object separately, the system finds the total field strength at a specific point and divides it between bodies in inverse proportion to their mass via the barycentre. This creates a perfect illusion of "opposing forces," although in reality, it is merely an optimized way of updating coordinates. Newton's third law is not the result of a struggle between two objects, but the basic code maintaining the integrity of the simulation grid.

This geometric model is ideal for virtual universes where complex matter properties are reserved for future updates. By eliminating the need for density calculations or particle distribution, it allows users to participate in world-building—creating stars and planets with physically accurate mechanics through a streamlined, efficient framework.